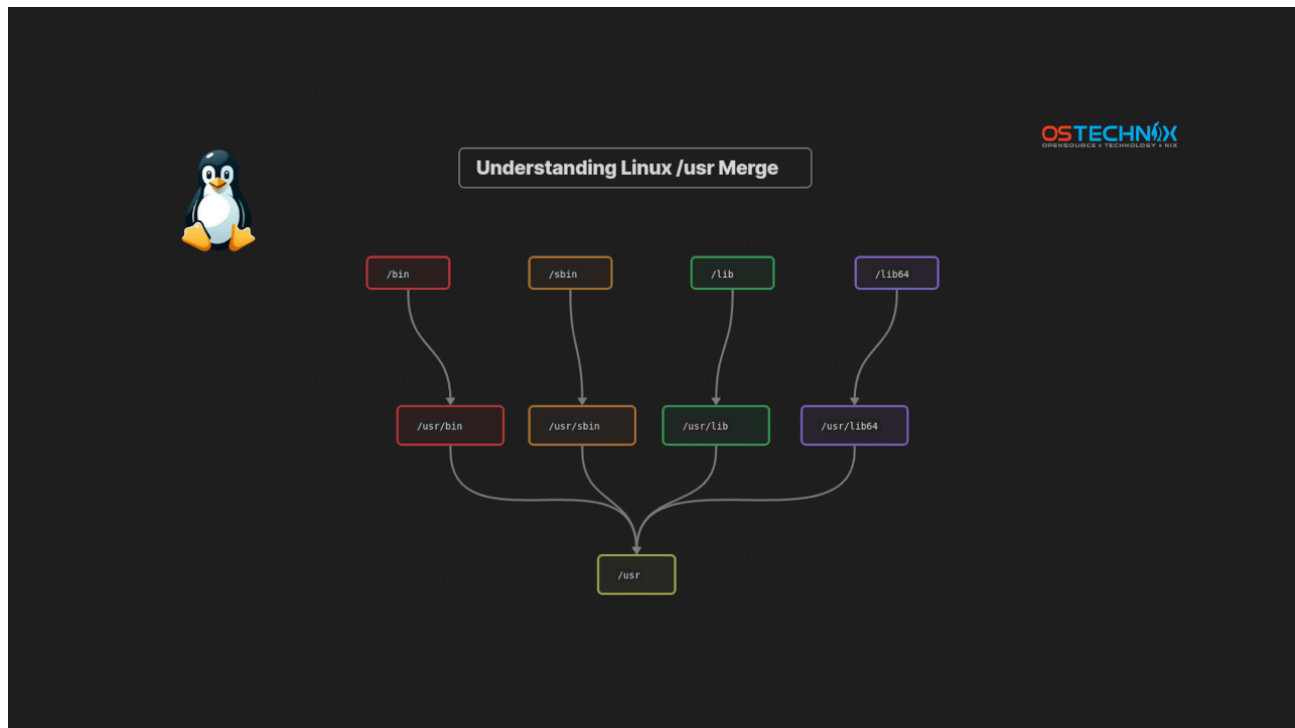


Understanding the Linux /usr Merge

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Today, we will discuss about the **"/usr merge"**, a filesystem reorganisation in Linux distributions where directories like /bin, /sbin, and /lib are relocated into their corresponding counterparts within /usr, with symbolic links created for backwards compatibility.

Why are Linux distributions doing this? Are there any real benefits of the /usr Merge? Let's find out

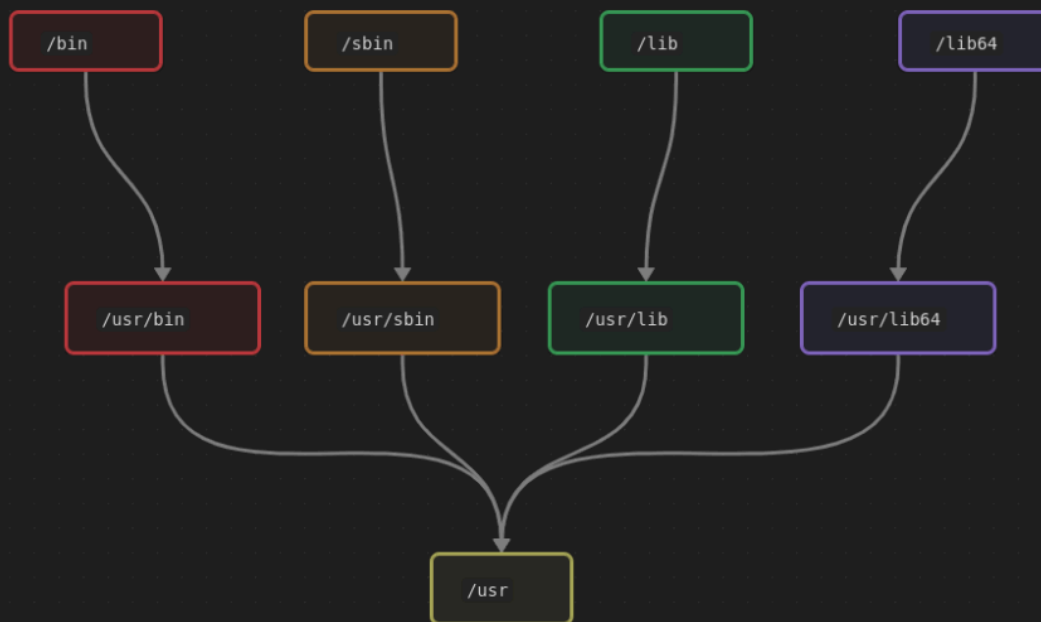
What is the /usr Merge, Anyway?

For a long time, important system files in the Linux filesystem were split between directories like /bin and /usr/bin, /sbin and /usr/sbin, /lib and /usr/lib, and /lib64 and /usr/lib64.

The **/usr Merge** is basically about putting all these files into just one set of folders under /usr. So, files from /bin go into /usr/bin, files from /sbin go into /usr/sbin, and so on.

Here is the visual representation of /usr merge:

Understanding Linux /usr Merge



Visual Representation of the Linux Usr Merge

To make sure old programs and scripts still work, [symbolic links](#) (symlinks) are created. These symlinks (shortcuts) point from the old places, like `/bin`, to the new places, like `/usr/bin`.

This way, if a program looks for something in `/bin`, the shortcut sends it to `/usr/bin` where the file now lives.

Why Are Linux Distributions Merging /usr?

This change might seem small, but it helps in several important ways.

1. Better Compatibility

Different versions of Unix or Linux distributions sometimes put the same programs in slightly different places. The /usr Merge helps make these locations the same by having programs available in both the old and new spots thanks to the symlinks.

This means scripts written for one Linux system might just work on another without needing changes, making things more compatible.

This merge has already been completed in some systems. [Fedora 17](#) successfully implemented this feature. **Oracle Solaris** also implemented the core part of the merge about 15 years ago and completed it with the introduction of Solaris 11.

2. Less Mess, More Simple

Having files spread across many top-level folders and duplicated in `/usr` makes things more complex than needed. The merge cleans up the file system layout, making it easier to understand where the main parts of the operating system are kept.

3. Easier for Developers

Developers often use tools that don't expect programs to be split between `/bin` and `/usr/bin`. The merge makes it simpler for them to build software for Linux.

It also helps Linux systems stay closer to how software is being built by developers everywhere.

4. New Cool Features

Putting most of the Linux operating system's parts into one main folder (`/usr`) allows for some neat tricks. For example, it makes it easier to:

- **Share** the Linux system files between different computers over a network or with lightweight 'guest' systems running on one computer. This means less data is needed on each individual computer.
- Take **snapshots** of the entire operating system easily and reliably.
- Make the important operating system files **read-only**. This can make the system safer and more stable. In fact, a big reason for the merge is to make having a read-only `/usr` work even better than before.

Clearing Up Confusion

There were old reasons for splitting files like this, mainly related to using different hard drives with different speeds. But those reasons don't apply much anymore because of how modern systems start up.

Some people worried the merge would break things or make them less compatible, but the opposite is true. It *increases* compatibility and is supported even with old setups like having `/usr` on a separate drive.

It's also not a new idea just for one Linux system; others have been doing it or moving towards it.

Even getting this change done can be tricky! For example, the Solus Linux team [shared their experience](#) trying the merge. They faced challenges making sure all the necessary updates happened in the right order, calling it a "Chicken and Egg problem".

They had to come up with clever new ways using special 'marker files' and scripts that run when the computer starts to make sure the merge happens safely and correctly for everyone.

This shows that while the *idea* is simple, the *process* can take careful planning!

TL;DR

/usr Merge is the **unification of core system directories** within the `/usr` file hierarchy in Linux and other Unix-like systems.

This involves merging the contents of `/bin`, `/sbin`, `/lib`, and `/lib64` into their respective counterparts in `/usr` (i.e., `/usr/bin`, `/usr/sbin`, `/usr/lib`, and `/usr/lib64`), and replacing the original directories with **symbolic links** pointing to the new locations in `/usr`.

The primary motivation for the `/usr` merge is to **improve compatibility** and **reduce system complexity**.

Fedora has successfully implemented this feature in version 17 and even other systems like Oracle Solaris have done this kind of merge.

It's not just limited to Fedora and Solaris. Multiple other Linux distributions have been "working in a similar direction" and are "in the process to implement the `/usr` merge".

Conclusion

The `/usr` merge makes Linux filesystem hierarchy easier to understand by organizing files in a more consistent way. It helps systems work better together, simplifies development, and supports new features like better security and easier file sharing.

Yum Extender NG: A Graphical Front-end For DNF5, DNF And YUM

Simplify Linux Package Management with Yum Extender NG, the user-friendly Graphical User Interface (GUI) for DNF5 Package Manager.

16.3K views 5 mins read

In this guide, we will explain what **Yum Extender NG** is and how to **install and use Yumex-ng in Linux** to install, update, remove and manage software packages graphically.

Introduction

DNF5 and its predecessors, such as **DNF** and **YUM**, are command-line package managers for RPM-based Linux distributions like Fedora, RHEL, CentOS Stream, AlmaLinux, and Rocky Linux.

However, for those who prefer a graphical user interface, there are a few GUI front-ends available for DNF5. These frontends provide a user-friendly interface for package management tasks, including installation, removal, and updates of software packages.

Yum Extender NG or **Yumex-ng** is one such front-end for DNF5, DNF and YUM, which provides an intuitive interface for managing packages in a user-friendly way.

What is Yum Extender NG?

Yum Extender NG (aka **Yumex-ng**) is a graphical user interface (GUI) for popular package managers like DNF5, DNF, and YUM, used in RPM-based Linux distributions. It offers an intuitive way to install, remove, and manage software packages on the system.

Yum Extender NG is an upgraded and rewritten version of the old Yum Extender (or Yumex) with a more modern look and feel, utilizing GTK4/Libadwaita. The "NG" in Yum Extender NG stands for "Next Gen."

Yumex-ng simplifies the process of installing, removing, upgrading, and updating packages with a simple yet fully functional GUI. It's designed for users who may not be comfortable using the command line interface of DNF5.

With Yumex-ng, users can browse and search for software packages, view detailed information about each package, and easily install or remove them with just a few clicks.

One notable feature of **Yumex-ng** is its ability to automatically handle dependencies. When a user selects a package to install, Yumex-ng will detect and install any required dependencies, saving time and effort compared to manually installing each package and its dependencies separately.

It also offers filtering and sorting options, enabling users to quickly find the software they need based on criteria such as package name, category, or repository. Users can customize the package list view by choosing which columns to display and sorting the packages in ascending or descending order.

Another useful feature of **Yumex-ng** is its support for multiple repositories. Users can easily add and manage additional software repositories, which is helpful if they need to install software not available in their system's default repositories.

It is highly customizable, with many options for tweaking the interface and application behavior. Users can adjust the font and color scheme, enable or disable features, and even customize the command-line options used by Yumex-ng when running in the background.

Yumex-ng is a free, open-source program written in **Python**. It can be used on any Linux distribution that supports DNF5. DNF and YUM package managers.

Install Yum Extender NG / Yumex-ng in Fedora Linux

I tested the following steps on latest Fedora 42 Workstation edition. I hope this will work on other Fedora versions as well.

Yum Extender NG, or Yumex-ng, can be installed via [Fedora Copr repository](#).

Add the [Yumex-ng copr repository](#) using command:

```
sudo dnf copr enable timlau/yumex-ng
```

Install Yumex using command:

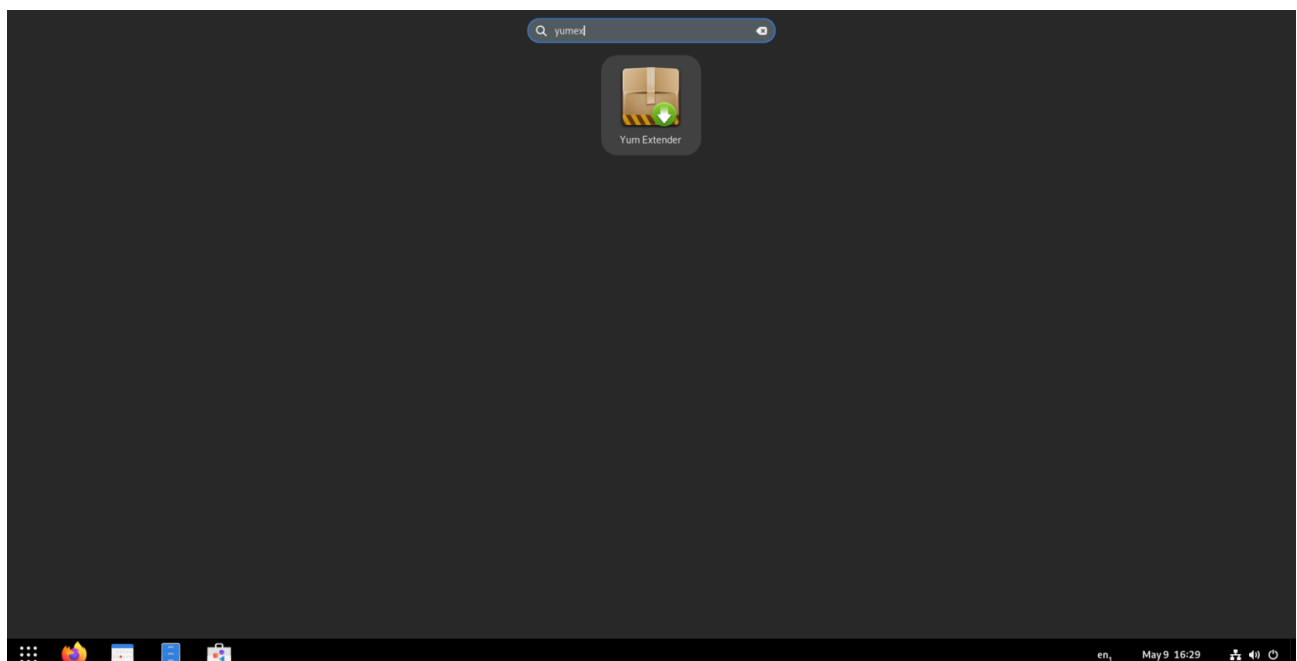
```
sudo dnf install yumex
```

If the old yumex-dnf package is installed, it will automatically remove it from your system.

How to Use Yum Extender NG?

Yum Extender NG usage is very simple and straight forward.

Launch Yumex from the dash or menu.



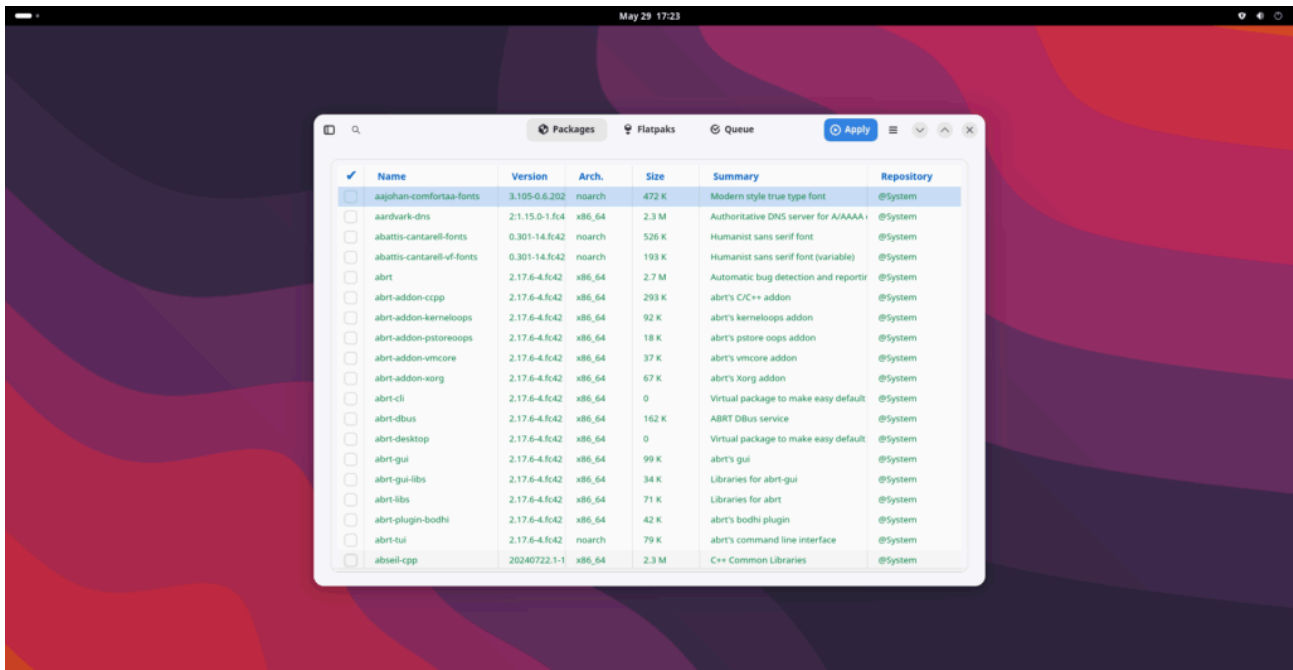
Launch Yum Extender NG

This will open the yumex-ng graphical user interface (GUI), allowing you to manage your system's packages.

You can also launch yumex-ng from the command line by entering the following command:

```
sudo yumex
```

Here is how the default interface of **Yumex-ng** looks like in Fedora 42.



Yum Extender NG Default Interface in Fedora 42 Linux

With yumex-ng installed and configured, you can now start using it to manage your system's packages. Browse through the available packages, select the ones you want to install or update, and click on the **'Apply'** button to make the changes.

Filter Packages

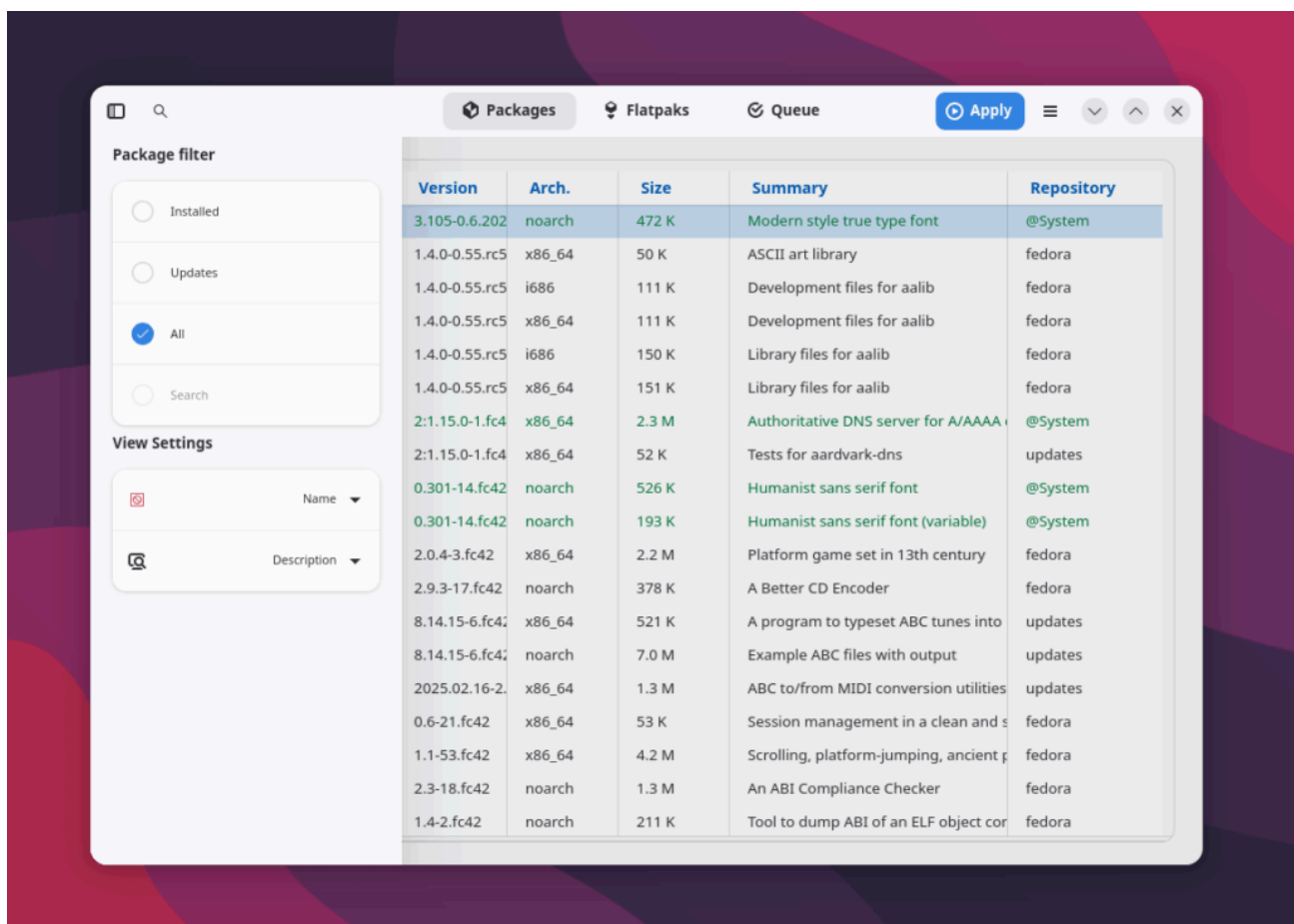
By default, Yumex-ng will show only the installed packages. The installed packages will be shown in green color.

If you want to show all other packages, click the **"Select Package filters"** icon on the top left of the Yumex-ng interface.



Select Package Filters in Yumex-ng

Now you can select either **Updates** to display the packages that need to be updated or **All** (both installed and not-installed) from filters section. I am going to list all packages.



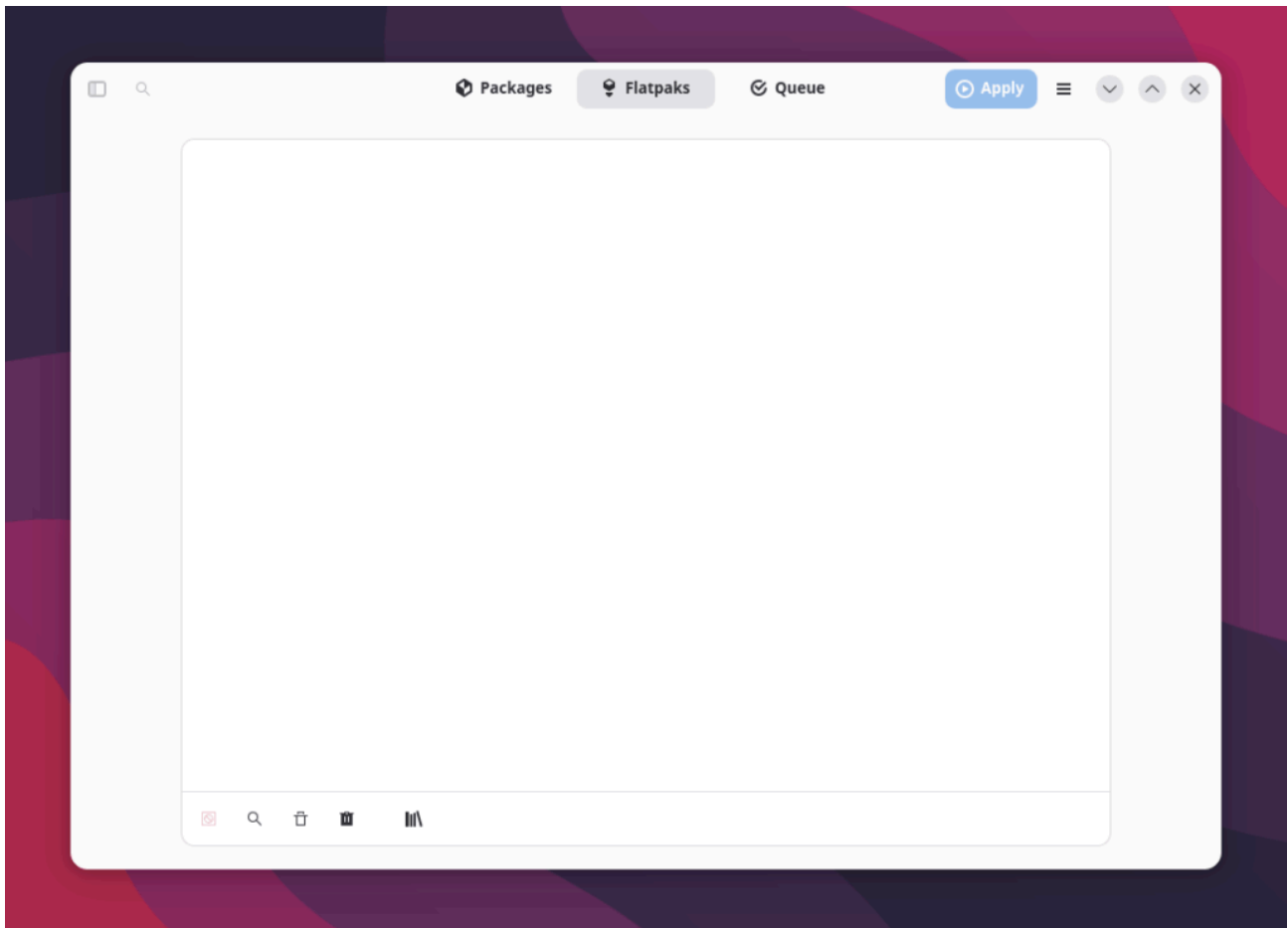
Show All Packages in Yumex-ng

Now choose any non-installed package and click the "Apply" button to install it.

Display Flatpaks

With the new version, you can also browse the installed flatpak applications as well.

To display flatpak apps, click the **"Flatpaks"** section on the top. If you have installed any flatpak applications, they will be shown here. If not, you will just see an empty screen.



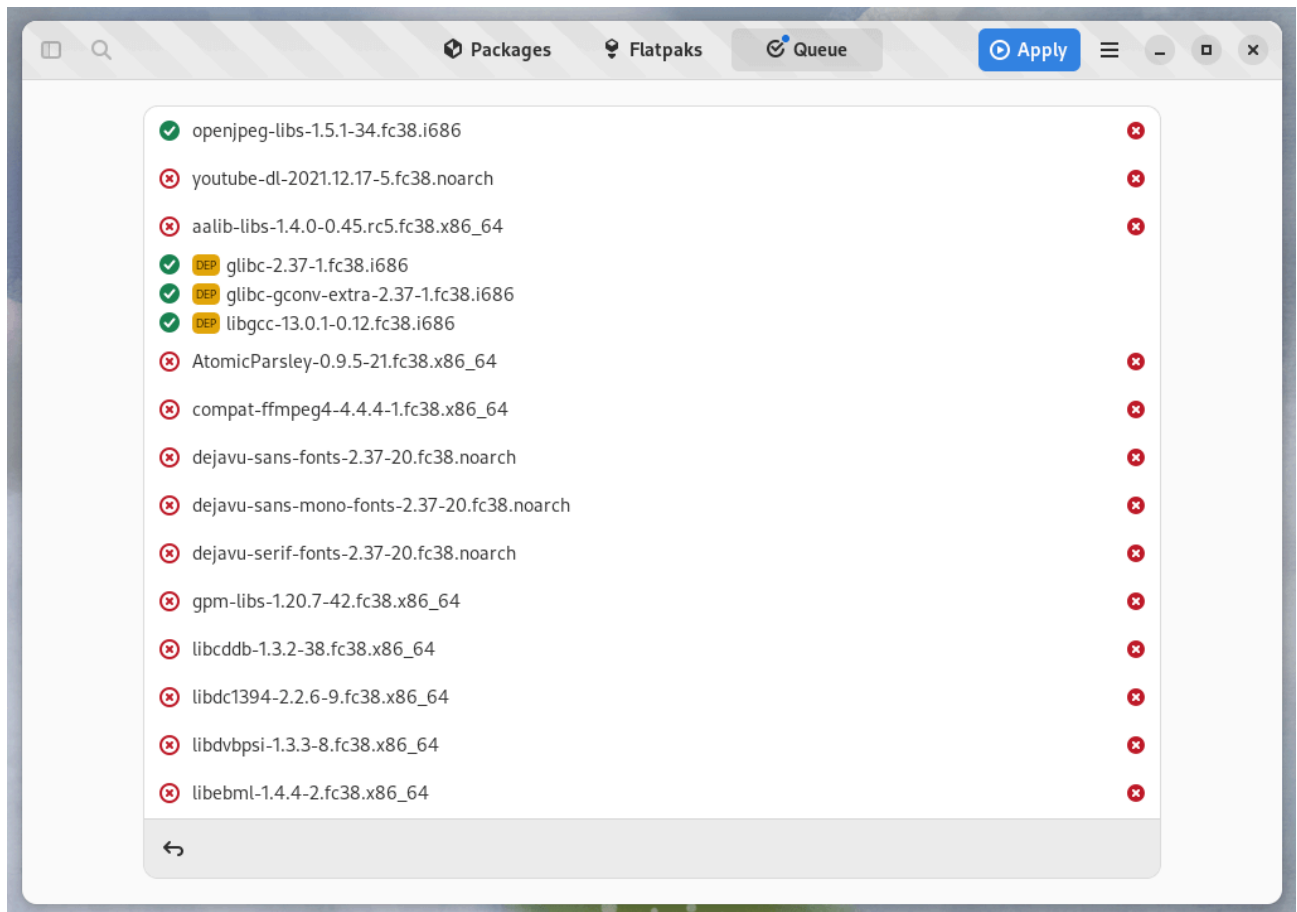
Show Flatpaks in Yumex-ng

To install new a flatpak app, click the **Search** button, choose or enter the name of the package to install, and follow the on-screen instructions.

Before installing flatpaks, you should have added a remote repository as described in the [Flatpak installation guide](#).

Queue Section

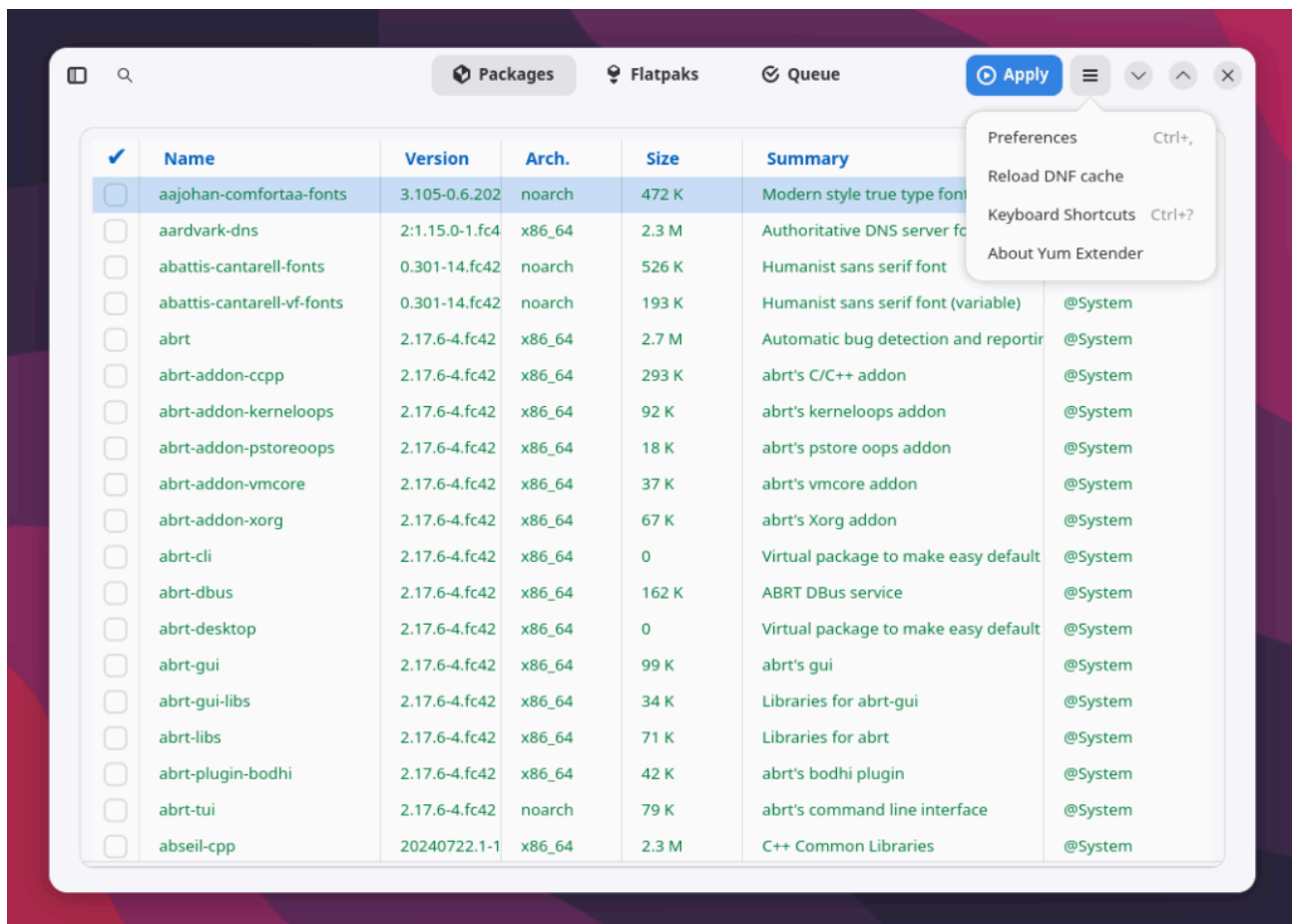
This section shows the list of applications marked for installation, updation and deletion.



Queue Section

View Repositories

To view the repositories, click the three horizontal bar icon and select **Preferences** option.



Open Preferences Tab

Navigate to the Repositories section, where you can enable or disable repositories.



[Repositories Section in Yum Extender NG](#)

Repositories Section in Yum Extender NG

Conclusion

Yumex-ng is a user-friendly package management tool that greatly simplifies the process of installing, removing, and managing software packages on RPM-based Linux systems. Whether you're new to Linux or an experienced user, Yumex-ng is worth exploring.

By following the steps in this guide, you can quickly install and configure Yumex-ng on your Linux system and start managing your packages with ease.